

Thyroid — Hypothyroidism (Hashimoto & Lab Traps)



1. Iodine deficiency worldwide

WHAT YOU SEE

Globe 'IODINE DEFICIENCY = #1 WORLDWIDE'

WHAT IT ENCODES

Iodine deficiency is the most common cause of hypothyroidism (and goiter) WORLDWIDE — iodine is the substrate for T3/T4, so deficiency limits synthesis, raises TSH, and causes goiter. In endemic deficiency it also causes endemic cretinism in newborns. Salt iodization is the public-health countermeasure.

BOARD TRAP

Calling Hashimoto's the #1 cause worldwide — wrong; Hashimoto's is #1 only in iodine-SUFFICIENT regions. Globally, iodine deficiency leads.

2. >95% autoimmune (Hashimoto)

WHAT YOU SEE

Banner '>95% AUTOIMMUNE'

WHAT IT ENCODES

In iodine-sufficient countries, the leading cause of hypothyroidism is Hashimoto's (chronic autoimmune lymphocytic) thyroiditis — T-cell-mediated destruction with lymphocytic infiltration and germinal centers (Hürthle/Askanazy cells on histology). It produces a firm, non-tender goiter and gradual gland failure; associated with HLA and other autoimmune diseases.

BOARD TRAP

Forgetting the geographic dependence — in iodine-sufficient areas autoimmune (Hashimoto's) dominates; worldwide it is iodine deficiency.

3. Anti-TPO antibody

WHAT YOU SEE

Doctor villager / antibody cue

WHAT IT ENCODES

Hashimoto's is marked by anti-thyroid peroxidase (anti-TPO) antibodies — the most sensitive marker — and anti-thyroglobulin antibodies. Anti-TPO positivity also predicts progression of subclinical hypothyroidism to overt disease and post-partum thyroiditis.

BOARD TRAP

Naming TRAb as the Hashimoto antibody — wrong; TRAb (TSI) is the Graves' stimulating antibody. Hashimoto's = anti-TPO / anti-thyroglobulin.

4. Hashimoto → lymphoma

WHAT YOU SEE

Grim reaper 'LYMPHOMA — RAPID MASS, ARISES FROM HASHIMOTO'S'

WHAT IT ENCODES

Long-standing Hashimoto's is the substrate for primary thyroid lymphoma (typically MALT or diffuse large B-cell), presenting as a rapidly enlarging neck mass in an older patient. Diagnosis is by biopsy; treatment is chemotherapy/radiation, NOT surgery.

BOARD TRAP

Assuming a rapidly growing mass in a Hashimoto patient is anaplastic carcinoma and rushing to surgery — biopsy first; lymphoma is treated medically and is far more curable.

5. APS-2 association

WHAT YOU SEE

'APS-2 = ADDISON + THYROID', Addison/T1DM/pernicious/vitiligo/celiac

WHAT IT ENCODES

Autoimmune polyglandular syndrome type 2 (Schmidt syndrome): autoimmune thyroid disease + Addison's disease ± type 1 diabetes, plus other organ-specific autoimmunity (pernicious anemia, vitiligo, celiac). Adult onset, polygenic/HLA-linked. Crucially, in a patient with both, treat Addison's (give glucocorticoid) BEFORE starting levothyroxine.

BOARD TRAP

Starting levothyroxine before recognizing/treating coexisting Addison's — wrong; accelerating metabolism with thyroid hormone in untreated adrenal insufficiency can precipitate adrenal crisis. Treat cortisol first.

6. BIOTIN TRAP

WHAT YOU SEE

'BIOTIN TRAP — STOP 2 DAYS, ↓TSH ↑T4 = FAKE hyperthyroid'

WHAT IT ENCODES

High-dose biotin (B7, common in hair/nail supplements) interferes with streptavidin-biotin immunoassays, producing a FALSE pattern of low TSH with high free T4/T3 — mimicking Graves' (and can also falsely lower TRAb). Stop biotin ~2–3 days (≥48 h) before testing and recheck.

BOARD TRAP

Treating a biotin-induced lab pattern as real hyperthyroidism — wrong and dangerous; the clue is a clinically euthyroid patient on supplements. Recheck off biotin before acting.

7. Delayed relaxation reflex

WHAT YOU SEE

'SLOW REFLEX RELAX / DELAYED RELAX'

WHAT IT ENCODES

The 'hung-up' (delayed-relaxation) deep tendon reflex — slow return phase, classically at the ankle — is a hallmark physical sign of hypothyroidism caused by slowed muscle contraction/relaxation kinetics.

BOARD TRAP

Mistaking the slow relaxation phase for a normal or hyperreflexic response — in hyperthyroidism reflexes are brisk/quick, the opposite finding.

8. Myxedema / periorbital edema

WHAT YOU SEE

'MYXEDEMA — puffed periorbital face, nonpitting leg edema'

WHAT IT ENCODES

Generalized myxedema: deposition of hydrophilic glycosaminoglycans (hyaluronic acid) in the dermis causes NON-pitting, doughy edema, periorbital puffiness, macroglossia, and a hoarse voice. Severe decompensation → myxedema coma (hypothermia, hypoventilation, hyponatremia, altered mental status) — treat with IV levothyroxine + stress-dose steroids.

BOARD TRAP

Confusing generalized myxedema of HYPOthyroidism with PRETIBIAL myxedema (a Graves'/HYPERthyroid dermopathy on the shins) — same word, opposite disease.

9. Menorrhagia

WHAT YOU SEE

'MENORRHAGIA' pale woman

WHAT IT ENCODES

Hypothyroidism causes menstrual disturbance — classically menorrhagia (and oligo-/anovulation, infertility) — partly via altered GnRH/prolactin dynamics (TRH can raise prolactin). Untreated hypothyroidism is a reversible cause of subfertility.

BOARD TRAP

Working up heavy menstrual bleeding without checking TSH — hypothyroidism is a treatable, easily missed cause.

10. Carpal tunnel

WHAT YOU SEE

'CARPAL TUNNEL'

WHAT IT ENCODES

Hypothyroidism is a classic REVERSIBLE cause of carpal tunnel syndrome — glycosaminoglycan deposition and soft-tissue swelling compress the median nerve in the carpal tunnel. Symptoms improve with thyroid replacement.

BOARD TRAP

Going straight to carpal tunnel surgery without checking for treatable systemic causes such as hypothyroidism (and pregnancy, diabetes, acromegaly).

11. ↑CPK & ↑lipids

WHAT YOU SEE

Lab board '↑CPK, ↑CHOLESTEROL/TG'

WHAT IT ENCODES

Hypothyroid metabolic labs: elevated creatine kinase (myopathy from slowed muscle metabolism) and hyperlipidemia — high LDL and triglycerides from reduced LDL-receptor expression and clearance. Lipids improve once euthyroid.

BOARD TRAP

Attributing an elevated CK to a statin myopathy or rhabdomyolysis without checking TSH, or starting lifelong lipid therapy before correcting the reversible hypothyroid dyslipidemia.

12. Hyponatremia & anemia

WHAT YOU SEE

'HYPONATREMIA, ANEMIA normo or macro'

WHAT IT ENCODES

Hypothyroidism causes hyponatremia (reduced free-water clearance, SIADH-like) and anemia that is typically normocytic, but can be macrocytic — and remember Hashimoto associates with pernicious anemia (B12 deficiency → true macrocytic) via APS. Microcytic anemia may coexist if menorrhagia causes iron deficiency.

BOARD TRAP

Chasing macrocytic anemia as purely B12/folate without considering hypothyroidism itself (and the Hashimoto–pernicious anemia association).

13. Drug causes (amiodarone, lithium)

WHAT YOU SEE

Pharmacy 'AMIODARONE, LITHIUM, IODINE EXCESS/DEFICIENCY'

WHAT IT ENCODES

Drug- and iatrogenic causes: amiodarone (iodine load → Wolff-Chaikoff-mediated hypothyroidism, type 1; can also cause thyrotoxicosis), lithium (blocks hormone release), iodine excess or deficiency, plus post-radioiodine, post-thyroidectomy, and external neck radiation. Also checkpoint inhibitors (immune thyroiditis).

BOARD TRAP

Forgetting amiodarone can cause EITHER hypo- OR hyperthyroidism — do not assume direction; the iodine load drives hypothyroidism in many patients but amiodarone-induced thyrotoxicosis also occurs.

14. Non-adherence detective

WHAT YOU SEE

'NON-ADHERENCE PATTERN — ↑TSH + NL T4'

WHAT IT ENCODES

A treated patient with persistently HIGH TSH but NORMAL/high free T4 suggests erratic adherence — the patient takes several missed pills just before the blood draw, normalizing T4 while the chronically under-replaced state keeps TSH elevated (long TSH lag). Confirms a behavioral, not pharmacologic, problem.

BOARD TRAP

Escalating the levothyroxine dose for this 'high TSH, normal T4' pattern — wrong; it reflects non-adherence (and the long TSH half-life), so increasing dose risks toxicity once the patient takes it consistently. Address adherence first.

15. Constipation

WHAT YOU SEE

'CONSTIPATION' figure

WHAT IT ENCODES

Slowed gut motility from thyroid hormone deficiency causes constipation — part of the global hypometabolic slowing (bradycardia, cold intolerance, fatigue, weight gain). It is the mirror image of the hyperdefecation/diarrhea of thyrotoxicosis.

BOARD TRAP

Treating chronic constipation symptomatically without checking TSH in a patient with other hypometabolic features.

16. Dry skin / hair loss

WHAT YOU SEE

'DRY SKIN / ALOPECIA'

WHAT IT ENCODES

Hypothyroidism produces dry, coarse, cool skin, brittle hair, and diffuse hair thinning including the lateral third of the eyebrows (Queen Anne's sign / madarosis) — from reduced skin metabolism, sweating, and sebum.

BOARD TRAP

Attributing the skin/hair changes to age or dermatologic disease alone and missing the reversible endocrine cause.

17. Pituitary hyperplasia mimic

WHAT YOU SEE

'PITUITARY HYPERPLASIA — MIMICS ADENOMA IN CHILDREN'

WHAT IT ENCODES

Severe long-standing primary hypothyroidism causes thyrotroph (and lactotroph) HYPERPLASIA from loss of negative feedback (high TRH/TSH drive), which can enlarge the pituitary and mimic an adenoma on MRI — and raise prolactin. The 'mass' SHRINKS with thyroid hormone replacement; it is not a true tumor.

BOARD TRAP

Sending a child/young patient with an enlarged pituitary from untreated hypothyroidism to pituitary surgery — wrong; treat the hypothyroidism and the pituitary regresses. Confirm TSH before imaging-driven decisions.
